

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Seventh Semester B.E. (Biotechnology) (Credit System) Degree Examinations**

January 2023

19BTC101 - ADVANCED PLANT TISSUE AND ORGAN CULTURE

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **Unit – I** and **Two full** questions from **Unit – II & Unit – III** each.

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	Anther/pollen is good source for the generation of haploid plants. Justify the statement with appropriate protocol.	10	L*2	1	1
	b)	Somatic hybridization techniques is an effective method of mass propagation and conservation of elite varieties. Substantiate the statement.	10	L2	1	1
2.	a)	What is organogenesis? Explain the types and factors influencing it.	10	L2	1	1
	b)	Write note on i) Endosperm culture ii) Somaclonal variants.	10	L2	1	1
		Unit – II				
3.	a)	Transgenic plants can be produced using gene transformation. Explain the protocol with example.	10	L2	2	2
	b)	Explain: i) Microprojectile bombardment ii) Site specific recombination.	10	L2	2	2
4.	a)	What is Binary plasmid vector? Explain its importance.	10	L2	2	2
	b)	Write note on: i) SCAR ii) Chloroplast transformation.	10	L2	2	1
5.	a)	Write a note on: i) Reporter genes ii) Gene silencing.	10	L2	2	1
	b)	Give a comparison between the advantages of Nuclear and Chloroplast genetic engineering.	10	L2	3	1
		Unit – III				
6.	a)	Elaborate the protocol for the production of draught resistant plants.	10	L2	4	2
	b)	Discuss the protocol for the induction of insect and pest resistant plants.	10	L2	5	2
7.	a)	Explain the method employed to induce delayed ripening of fruits.	10	L2	4	2
	b)	Give an account of plantibodies.	10	L2	5	2
8.	a)	What is metabolic engineering? Explain its applications.	10	L2	4	2
	b)	Herbicide resistant crops can be produced using genetic engineering approach. Justify with an example.	10	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
